Quantitative Methods 1

Chapter 1: Organising and presenting data

Detailed solutions

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Exercise 1 — A survey of retail outlets

A chain operates 180 shops and randomly selects 60 from its complete list. For each store it records the main sales channel (physical store, website or mobile application), weekly product returns, daily turnover and an ordered service-quality rating.

1. Identify the population, sample and statistical unit.
2. Classify all four variables and choose a suitable graph for each.
3. For which variables do cumulative frequencies make sense?
4. Describe two possible sources of selection bias.

Fully worked correction

No frequency table can be computed because this exercise provides variables but no observed category counts. The starting statistical inventory is therefore:

Variable	Values / unit	Type	Graph
Sales channel	in-store, website, app	Nominal	Separate bars
Weekly returns	$0, 1, 2, \ldots$	Discrete numerical	Sticks
Daily turnover	EUR/day	Continuous numerical	Histogram
Service rating	low / medium / high	Ordinal	Ordered bars

1. Population, sample, unit

The statistical population comprises the 180 stores operated by the chain. The observed sample comprises 60 surveyed shops. One shop corresponds to one row of data and is the statistical unit. The number 60 is a sample size, not the number of possible values of a variable.

2. Classification and suitable displays

A sales channel has no intrinsic ranking; sorting categories alphabetically does not create statistical order. Returns are nonnegative integer counts. Revenue is measured on a quantitative scale even when rounded to cents. A rating is ordered but differences between levels are not inherently equal. A histogram is appropriate for revenues only after defining intervals and their widths.

3. Cumulative frequencies

An ascending cumulative frequency at an ordered category/value is the sum of the frequencies up to that category/value. Therefore it can be defined for returns, revenues and ratings, but not meaningfully for unordered sales channels. If counts were available, the generic formula would be:

$$F_i = \sum_{j=1}^{i} f_j = \frac{\sum_{j=1}^{i} n_j}{N}.$$

4. Two sources of bias

Example 1: the poorest-performing stores might refuse the survey (nonresponse bias). Example 2: collecting observations only from one region or one opening period omits other shop types (coverage/selection bias). Increasing 60 to 600 observations from the same biased mechanism does not remove that bias.

Managerial interpretation and consistency check

The survey covers $60/180 = 1/3 = 33.33\%$ of the chain. Sampling a large share does not by itself eliminate bias: the probability of inclusion, the completeness of the sampling frame and non-response matter. A store is the statistical unit, not a customer or a transaction. No channel-frequency table may be calculated because the exercise gives variable definitions, not shop-level category counts. A main channel is a sales route, not a product range. Cumulative proportions are meaningful only where an ordered sequence has a substantive interpretation.

Exercise 2 — Product returns: discrete distribution

Weekly product returns were counted in 60 shops:

Returns x_i	0	1	2	3	4	5
Shops n_i	5	11	19	14	8	3

1. Check the total number of shops.
2. Build the full table of counts, frequencies, ascending and descending cumulative counts and frequencies.
3. Draw a stick diagram.
4. Find the proportions with at most two and at least three returns.

Fully worked correction

First construct the complete statistical table; all subsequent calculations and graphs must use these same values.

x_i	n_i	f_i	$100f_i$	N_i^+	N_i^-	F_i^+	F_i^-
0	5	5/60=0.0833	8.33	5	60	0.0833	1.0000
1	11	11/60=0.1833	18.33	16	55	0.2667	0.9167
2	19	19/60=0.3167	31.67	35	44	0.5833	0.7333
3	14	14/60=0.2333	23.33	49	25	0.8167	0.4167
4	8	8/60=0.1333	13.33	57	11	0.9500	0.1833
5	3	3/60=0.0500	5.00	60	3	1.0000	0.0500
Total	60	1.0000	100.00	–	–	–	–

1. Check total count

$$N = 5 + 11 + 19 + 14 + 8 + 3 = 60.$$

2. How every column is computed

For each value: relative frequency = count / 60; increasing cumulative count = this count plus all preceding counts; decreasing cumulative count = this count plus all following counts. The complete arithmetic is:

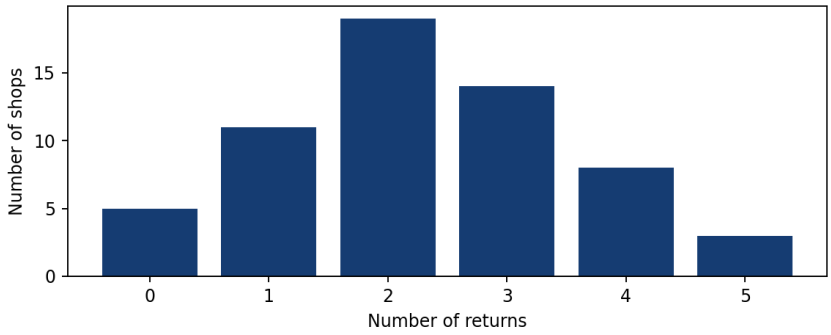
$$x = 0: \quad f_0 = 5/60 = 0.0833; \quad N_0^+ = 5 = 5; \quad N_0^- = 5 + 11 + 19 + 14 + 8 + 3 = 60.$$
$$x = 1: \quad f_1 = 11/60 = 0.1833; \quad N_1^+ = 5 + 11 = 16; \quad N_1^- = 11 + 19 + 14 + 8 + 3 = 55.$$
$$x = 2: \quad f_2 = 19/60 = 0.3167; \quad N_2^+ = 5 + 11 + 19 = 35; \quad N_2^- = 19 + 14 + 8 + 3 = 44.$$
$$x = 3: \quad f_3 = 14/60 = 0.2333; \quad N_3^+ = 5 + 11 + 19 + 14 = 49; \quad N_3^- = 14 + 8 + 3 = 25.$$
$$x = 4: \quad f_4 = 8/60 = 0.1333; \quad N_4^+ = 5 + 11 + 19 + 14 + 8 = 57; \quad N_4^- = 8 + 3 = 11.$$
$$x = 5: \quad f_5 = 3/60 = 0.0500; \quad N_5^+ = 5 + 11 + 19 + 14 + 8 + 3 = 60; \quad N_5^- = 3 = 3.$$

Divide each cumulative count by 60. For example:

$$F_2^+ = 35/60 = 0.5833 = 58.33\%, \qquad F_3^- = 25/60 = 0.4167 = 41.67\%.$$

3. Construct the bar/stick chart

Horizontal coordinates are 0, 1, 2, 3, 4, 5; the corresponding vertical heights are 5, 11, 19, 14, 8 and 3. The most frequent count is two returns (19 shops). Each stick is located exactly on its integer value; the heights are not joined to make a continuous curve.



4. Probabilities and complements

$$P(X \le 2) = \frac{5+11+19}{60} = \frac{35}{60} = 0.5833 = 58.33\%.$$
$$P(X \ge 3) = \frac{14+8+3}{60} = \frac{25}{60} = 0.4167 = 41.67\%.$$
$$35 + 25 = 60, \qquad 58.33\% + 41.67\% = 100\%.$$

Worked column checks and business reading

The frequency of two returns is $f_2 = 19/60 = 0.316667$, or 31.67%. The increasing cumulative count at two returns is $5+11+19 = 35$ and the corresponding frequency is $35/60 = 58.33\%$. The decreasing cumulative count at three returns is $14+8+3 = 25$, corresponding to $25/60 = 41.67\%$. The two groups are mutually exclusive and exhaustive because $35 + 25 = 60$ and $58.33 + 41.67 = 100\%$ before rounding. Managers should not equate the absolute number of returned products with a return rate: stores may have very different numbers of sales.

Exercise 3 — Price survey: unequal class widths

A firm surveys 150 consumers about the maximum price they would pay for a portable solar charger. Classes are left-closed and right-open.

Class (euros)	12–18	18–24	24–30	30–36	36–45	45–60	60–75
Count n_i	6	14	28	42	30	20	10

1. Calculate class widths, relative frequencies and count densities.
2. Construct a correct histogram and both cumulative-count curves.
3. Find the shares willing to pay less than 36 euros and at least 36 euros.
4. Explain why raw class counts cannot be used as histogram heights.

Fully worked correction

Start with a complete table. Classes are left-closed and right-open, so 36 belongs to [36,45[, not [30,36[.

Class	n_i	a_i	f_i	Decimal f_i	d_i	N_i^+	N_i^-
[12,18[	6	6	6/150	0.0400	1.0000	6	150
[18,24[	14	6	14/150	0.0933	2.3333	20	144
[24,30[	28	6	28/150	0.1867	4.6667	48	130
[30,36[	42	6	42/150	0.2800	7.0000	90	102
[36,45[	30	9	30/150	0.2000	3.3333	120	60
[45,60[	20	15	20/150	0.1333	1.3333	140	30
[60,75[	10	15	10/150	0.0667	0.6667	150	10
Total	150	–	–	1.0000	–	–	–

1. Class widths, frequencies, densities

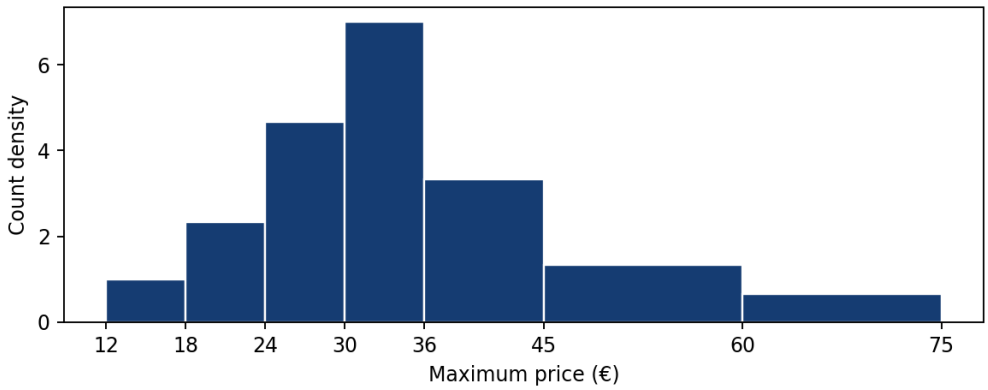
$$\sum n_i = 6+14+28+42+30+20+10 = 150, \quad a_i = b_i - a_i^{(L)}, \quad f_i = n_i/150, \quad d_i = n_i/\text{width}_i.$$

$$\begin{aligned}
[12,18[: &\quad a = 18-12 = 6, \quad f = 6/150 = 0.0400, \quad d = 6/6 = 1.0000.\\
[18,24[: &\quad a = 24-18 = 6, \quad f = 14/150 = 0.0933, \quad d = 14/6 = 2.3333.\\
[24,30[: &\quad a = 30-24 = 6, \quad f = 28/150 = 0.1867, \quad d = 28/6 = 4.6667.\\
[30,36[: &\quad a = 36-30 = 6, \quad f = 42/150 = 0.2800, \quad d = 42/6 = 7.0000.\\
[36,45[: &\quad a = 45-36 = 9, \quad f = 30/150 = 0.2000, \quad d = 30/9 = 3.3333.\\
[45,60[: &\quad a = 60-45 = 15, \quad f = 20/150 = 0.1333, \quad d = 20/15 = 1.3333.\\
[60,75[: &\quad a = 75-60 = 15, \quad f = 10/150 = 0.0667, \quad d = 10/15 = 0.6667.
\end{aligned}$$

2. Histogram: area, not height, represents counts

Draw adjacent rectangles with the exact interval as base, and density as height. Check for example:

$$[30, 36[: \quad \text{area} = 6 \times 7 = 42; \quad [45, 60[: \quad 15 \times (20/15) = 20.$$



3. Ascending and descending cumulative curves

Ascending cumulative counts are obtained at right boundaries, starting from 0 at the first lower boundary. Descending counts are obtained at left boundaries, starting at 150. Calculate each explicitly:

$$[12, 18[: \quad ECC = 6 = 6; \quad ECD = 6 + 14 + 28 + 42 + 30 + 20 + 10 = 150.$$

$$[18, 24[: \quad ECC = 6 + 14 = 20; \quad ECD = 14 + 28 + 42 + 30 + 20 + 10 = 144.$$

$$[24, 30[: \quad ECC = 6 + 14 + 28 = 48; \quad ECD = 28 + 42 + 30 + 20 + 10 = 130.$$

$$[30, 36[: \quad ECC = 6 + 14 + 28 + 42 = 90; \quad ECD = 42 + 30 + 20 + 10 = 102.$$

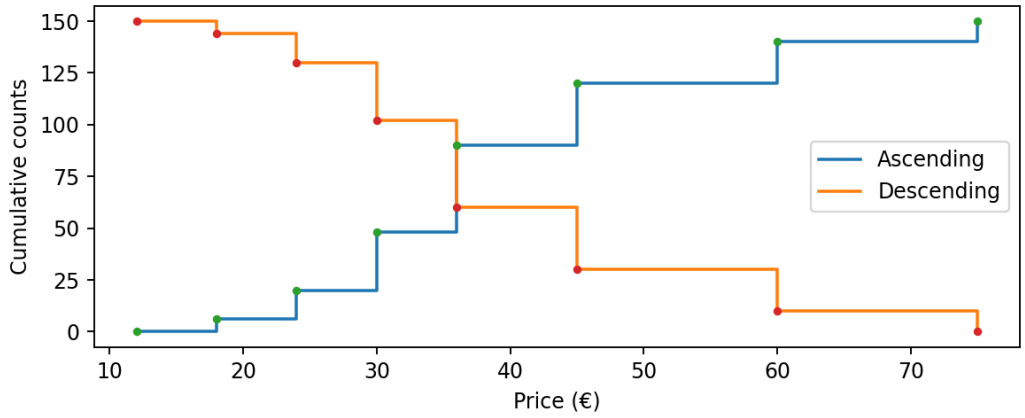
$$[36, 45[: \quad ECC = 6 + 14 + 28 + 42 + 30 = 120; \quad ECD = 30 + 20 + 10 = 60.$$

$$[45, 60[: \quad ECC = 6 + 14 + 28 + 42 + 30 + 20 = 140; \quad ECD = 20 + 10 = 30.$$

$$[60, 75[: \quad ECC = 6 + 14 + 28 + 42 + 30 + 20 + 10 = 150; \quad ECD = 10 = 10.$$

Coordinate table for drawing (EUR, customers):

Boundary	12	18	24	30	36	45	60	75
ECC	0	6	20	48	90	120	140	150
ECD	150	144	130	102	60	30	10	0



4. Price-threshold shares

$$P(X < 36) = \frac{6 + 14 + 28 + 42}{150} = \frac{90}{150} = 0.60 = 60\%.$$
$$P(X \geq 36) = \frac{30 + 20 + 10}{150} = \frac{60}{150} = 0.40 = 40\%.$$

These are exact relative frequencies for the grouped thresholds; no interpolation is required at a class boundary.

How to verify histogram bars and interpret the threshold

For the interval $[30, 36[$, width $a = 36 - 30 = 6$, count $n = 42$, density $d = 42/6 = 7$ customers per euro, and rectangle area $6 \times 7 = 42$. For $[36, 45[$, width 9, count 30, density $30/9 = 3.3333$ and area $9 \times (30/9) = 30$. The second class must not have a height of 30 while the first has a height of 42: that would confuse area and height when widths differ. Below EUR 36 there are $6+14+28+42 = 90$ consumers, hence $90/150 = 60\%$. At least EUR 36 there are $30+20+10 = 60$, hence $60/150 = 40\%$. The latter group can afford the listed price of EUR 36 only under the stated maximum-price criterion; observed purchases are not implied.